

Project Book Shop Management System

Tech Stack: C++ MySQL

Course: DBMS

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1. Introduction

The Book Shop Management System is a console-based desktop application developed using C++

and MySQL, designed to operate efficiently on the Windows operating system. The system

automates daily book shop operations such as employee management, member registration, and

record maintenance using a relational database.

2. Objectives

• Automate book shop operations

• Reduce manual record keeping

• Maintain employee and member details

• Provide fast search and update functionality

• Ensure data persistence using MySQL

3. Technologies Used

Programming Language: C++

Database: MySQL

Operating System: Windows

Compiler: MinGW g++ / Visual Studio

API: MySQL C API

4. System Architecture

The system follows a three-tier architecture consisting of Presentation Layer (Console UI),

Application Logic Layer (C++ classes and functions), and Database Layer (MySQL). This design

ensures modularity and maintainability.

5. Features

• Employee management

• Member registration

• Search and update records

• Manager assignment

• Salary updates

• Secure database connectivity

6. Database Design

The database includes employees, members, and managers tables. Each table uses a primary key

and relational structure to maintain data integrity.

7. Windows Compatibility Enhancements

Windows-specific libraries such as windows.h and conio.h were used. Screen clearing was handled

using system("cls"), delays using Sleep(), and MySQL Connector/C was configured for Windows

compilation.

8. Advantages

• Windows compatible

• Fast execution

• Easy maintenance

• Scalable design

9. Limitations

• Console-based interface

• No authentication system

• No encryption

10. Future Enhancements

• GUI using Qt

• Role-based access control

• Password encryption

• Web-based frontend

• Logging and audit trails

11. Conclusion

The Book Shop Management System demonstrates effective use of C++ and MySQL on the

Windows platform. It highlights database integration, system programming concepts, and modular

software design suitable for academic and small business use.